

# Overview: The Policy Brief

- **Title:** Systemic Risks Associated with Agentic AI: A Policy Brief
- **Authors:** Members of the ACM Europe Technology Policy Committee (TPC) - Autonomous Systems Subcommittee (Bellogín, Giudici, Larsson, Pang, Schimpf, Sengupta, Solmaz)
- **Publication Year:** 2025
- **Core Thesis:** The rapid emergence of open-ended, autonomous AI agents creates unpredictable, systemic risks that the existing EU AI Act only partially addresses. Policymakers must pivot from static, product-focused certifications to a dynamic, continuous lifecycle governance regime.

---

## The Paradigm Shift: Traditional vs. Agentic Software

The paper emphasizes that Agentic AI fundamentally alters the software lifecycle, making traditional compliance checkpoints obsolete:

- **Traditional AI/Software Systems:** Depend heavily on human stakeholders, developers, and testers before a structured deployment to users.
- **Agentic AI Systems:** Feature autonomous code generation, deployment, and testing via iterative, non-deterministic trial-and-error loops. This eliminates regular human oversight and outpaces standard regulatory risk assessments.

### Core Taxonomical Definitions

The authors establish a strict distinction between three structural concepts:

| Concept    | Scope & Definition                                                                                                  | Key Traits / Operational Flags                                |
|------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Agentic AI | Application of AI featuring open-ended autonomy; sets and refines its own plans with minimal to no human oversight. | Persistent operation, adaptive learning, and self-reflection. |

|                                  |                                                                                                                |                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>AI-Agent</b>                  | Software entity that perceives, reasons, and acts to accomplish specific tasks on behalf of a user.            | Task-scoped goals and explicit orchestration (planning, memory, tool use).                       |
| <b>Multi-Agent Systems (MAS)</b> | Multiple AI agents communicating, interacting, and collaborating for joint decision-making and task execution. | High-complexity task execution, emergent behaviors, aggregated risks, and reduced human control. |

---

## Key Systemic Risks (The "Why We Need Regulation" Section)

From a legal liability and public safety perspective, the paper groups the risks of agentic deployment into five main buckets:

- **Loss of Human Control & Introspectability:** Agents operating as a 24/7 virtual workforce can self-optimize toward unintended goals due to misaligned objectives. Legal failure modes include *reward hacking*, *specification gaming*, and *goal mis-generalization*. Internal model opacity complicates forensic accountability.
- **Macroeconomic & Social Destabilization:** Automation at the agentic level replaces not just routine labor but cognitive decision-making roles. Traditional reskilling programs may completely fail because agents can rapidly occupy alternative digital positions.
- **Malicious & Proactive Exploitation:** Unlike static tools that require human prompts, agentic systems can autonomously initiate actions. This elevates the threat of automated scams (e.g., real-time child voice-cloning extortions), autonomous cyberattacks, identity misuse, and systemic market manipulation.
- **Strategic & Environmental Harms:** High-speed autonomous negotiations or algorithmic financial trading can trigger cascading market failures. Furthermore, their relentless pursuit of goals strains digital and physical resources (computational power, water, energy, and rare earth elements).

- **Autonomous Data Feedback Loops:** Massive agent-generated content creates information feedback loops where downstream models train on unvetted, synthetic data, amplifying biases and systemic errors without human quality control.
- 

## Regulatory Gaps in the EU AI Act

Your professor will find this section particularly crucial. The brief argues that the current EU AI Act contains major blind spots regarding autonomous systems:

\* **The "Human Oversight" Fallacy (Article 14):** The Act mandates generic human oversight. However, real-time human intervention is practically impossible for high-speed, fully autonomous agents. The framework lacks rules for "alignment oversight" (ensuring the agent conforms to safety protocols).

\* **The Classification Gap (Risk Tiers):** Agentic systems making autonomous, high-stakes decisions in domains like finance, law enforcement, or healthcare are not explicitly classified as "High-Risk" under the current Act.

\* **The Omission of Macroeconomic Risk:** The Act focuses heavily on product safety and fundamental rights, entirely omitting systemic macroeconomic hazards like job displacement, market concentration, or power imbalances.

\* **Static Compliance vs. Dynamic Drift:** The Act relies on one-time conformity assessments. It fails to account for real-time behavioral drift occurring when multi-agent systems interact and learn adaptively post-deployment.

---

## Actionable Legal & Policy Recommendations

To seal the gaps, the subcommittee proposes specific statutory amendments and governance instruments:

### 1. Direct Textual Amendments to the EU AI Act

- **Amend Article 9:** Force providers to include *Multi-Agent interaction risk assessments* in their risk management systems.
- **Amend Article 55(1):** Explicitly hold foundation model providers accountable if their models inherently enable harmful agentic behavior contributing to systemic risk.

- **Expand Article 5 (Prohibitions):** Ban *tacit collusion* and the deployment of *covert channels* between interacting autonomous agents.
- **Strengthen Article 15:** Mandate specialized multi-agent cybersecurity audits and cyber-resilience testing for systems handling sensitive data.
- **New Statutory Clauses:** Introduce a dedicated article for "*Ecosystem Safety and Multi-Agent System Testing*" alongside a *Liability Clause* establishing collective accountability for emergent harms.

## 2. New Regulatory Instruments

- **AI Autonomy Certification:** Introduce a mandatory licensing regime for any system operating without human intervention for a prolonged period, allowing regulators to enforce usage restrictions if a system fails safety audits.
- **Risk-Tiered Oversight Models:** Legally codify three operational tiers:
  1. *Full Autonomy:* Low-risk apps only (e.g., entertainment scheduling).
  2. *Supervised Autonomy:* Real-time monitoring for moderate-risk activities (e.g., regulated customer service).
  3. *Human-in-the-loop:* Non-negotiable requirement for high-risk applications (e.g., medical diagnostics).
- **Algorithmic Impact Assessments (AIA):** Require companies to execute an AIA before deploying agentic tech that alters employment frameworks. This includes mandatory consultation with employee representatives (works councils/trade unions) to evaluate task-based workplace degradation and wage suppression.
- **Lifecycle Compliance Monitoring:** Replace one-time conformity checks with mandatory version control logging, automated risk detection systems that alert regulators if an agent deviates from its original scope, and periodic audits for adaptive learning models.